Bryan Qi

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Education

University of California, Los Angeles — Los Angeles, CA
B.S. Computer Science and Engineering, 3.84/4.00 GPA — *Jun 2028*

Experience

Software Engineering Intern — Jul 2025 – Sep 2025
Swifter Fans — *Denville, NJ*
- Built internal **Python** and **Flask** full-stack web app generating HVLS fan layouts from client needs and ASHRAE-55 targets
- Implemented dynamic client-ready proposal generation using **python-docx** with **Matplotlib** visuals, producing reports in **< 3s**
- Employed **Flask-Session** session management to persist layout selections across requests and handle file generation/downloads
- Consolidated manual **2-step** process, reducing response times to leads from **days to minutes**, and serving **500** avg. monthly visits
- Leveraged **CadQuery** to construct **3D STEP models** of fan placements in client warehouses for future CFD analysis
- Experimented with **Streamlit** and **tkinter** dashboards before refactoring into **Flask** and deploying on **PythonAnywhere**

Rocket Data Visualization Engineer — Sep 2025 – Jun 2026
UCLA Rocket Project, Prometheus — *Los Angeles, CA*
- Rebuilt legacy telemetry pipeline with **C++** for **ESP-32** serial data packaging and **Python WebSockets** using **requests** library
- Migrated data visualization stack to **Telegraf**, **InfluxDB**, and **Grafana** for live pressure transducer and load cell monitoring at **1 kHz**
- Collaborated with other software members to support the propulsion and electronics teams during midnight cold-flow tests

Software Developer — Dec 2025 – Jun 2026
Bruin Software Engineers — *Los Angeles, CA*
- Deployed **Python** automation on **Google Cloud Platform** to verify data across spreadsheets in real-time using **Google Sheets API**
- Automated notification emails using **Gmail API** and **schedule** library to schedule batch API updates and emails to optimize compute

Materials Science Research Intern — Jun 2024 – Jul 2024
Belcher Lab @ MIT Koch Institute for Integrative Cancer Research — *Cambridge, MA*
- Benchmarked Li-air batteries with **microporous carbon nanofiber anodes** produced by **E. coli** and **bacteriophage biotemplating**
- Supported the preparation and clean-up of solutions for accelerated growth of artificial lab-grown cellulose for **PFAS** filter research
- Shadowed multiple projects, learning **gas physisorption**, **transmission electron microscopy**, and **spectrophotometry** techniques

Quantitative Finance Lecturer & Track Lead — Jun 2026 – Present
Bruin Software Engineers — *Los Angeles, CA*
- Designed a **4-week** introduction to **quantitative finance** and **algorithmic trading** with guest lecturers from **Optiver** and **Barclays**
- Contacted and met with engineer and trader from **Citadel Securities** and **Jane Street** to curate list of impactful topics and projects
- Wrote lectures on **market microstructure**, **momentum trading**, **mean reversion**, **abritrage**, **market making**, **HFT**, **Sharpe ratio**
- Built standardized backtester system in **Python** for easy strategy testing and evaluating for students using **Bitcoin Cash (BCH)** data
- Organized competition to develop strategies for **Kalshi** prediction markets and backtest student-tuned **BCH** strategies on unseen data

Projects

Voices Around Us – Hackathon Winner | *React Native, Supabase, PostgreSQL, Git, Expo, OpenAI API, OpenAI Whisper, Figma*
- Won **Best in Storytelling** at ACM's HOTH XIII among **94** competitors for developing a community sharing and discovery platform
- Developed **location-based mobile storytelling platform** pinning user-generated audio and text stories to geographic locations
- Integrated **OpenAI Whisper** to automatically transcribe user audio stories into text, **improving accessibility** and **content searching**
- Implemented **location-based content discovery system** to foster understanding through glimpses into past lived experiences nearby

Limit Order Book and Matching Engine | *C++20, CMake, Google Benchmark, Google Test*
- Designed a **price-time priority** order book and matching engine in **C++20** from scratch to simulate matching market and limit orders
- Built differential testing pipeline to verify trades and book states against reference implementation across **25,000+** randomized orders

SecondLook – Hackathon Project | *Next.js, React, Supabase, Clerk Auth, Google Gemini API, Vercel, Tailwind CSS*
- Deployed recruiter-triage platform employing **Gemini**-powered applicant **tracking**, **ranking**, and **insights** across job descriptions
- Contacted and **interviewed T-Mobile recruiter** for insights on common pain points and daily challenges for recruiting teams
- Crafted slide deck and **pitched product to 5 Google engineers** at PLV for feedback and judging

AI Portfolio Assistant with LLM-Judge Loop | *Python, OpenAI API, Pydantic, Gradio*
- Built **personal assistant** that answers recruiter-style questions using **resume data**, **cover letter**, and **LinkedIn profile**
- Implemented **LLM-as-a-judge** loop to verify responses against criteria and triggering regeneration with feedback until it passes

Skills and Certificates

Languages: Python, C/C++, SQL (PostgreSQL), JavaScript, TypeScript, HTML/CSS
Frameworks: React.js, React Native, Node.js, Flask, Next.js
Developer Tools: Git, Google Benchmark, CMake, Expo, Jupyter Notebook, VS Code, Xcode, Claude Code, Linux
Certificates: AWS PostgreSQL for Developers, Anthropic Claude Code 101
Data and APIs: REST API, JSON, Google Cloud, Gemini API, OpenAI API